Software Requirements Specification

for

Süprüz

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version	Who
Template Creation	07/03/26	Start of the project.	0.0.1	Beyza
Introduction & Conventions	10/03/26	Added introduction, purpose and document conventions.	0.0.2	Hayrunnisa
Audience, Scope & References	11/03/26	Added intended audience, project scope and references.	0.0.3	Hayrunnisa
Overall Description	19/03/26	To define the Süprüz with its details.	0.0.4	Emre
Nonfunctional Requirements	22/03/26	Defined performance, safety, security, software quality attributes, and business rules.	0.0.5	Enes
System Features	23/03/26	Added functional requirements, stimulus/response sequences and priority ratings for all six system features.	0.0.6	Beyza
User Interfaces	24/03/26	To provide a detailed, screen-by-screen breakdown of all app interfaces, clarifying layouts and user interactions.	0.0.7	Umut
External Interfaces	24/03/26	To explicitly define and map all hardware, software, and communication protocols to their corresponding system requirements.	0.0.8	Umut
Appendix A&B	24/03/26	Added Glossary and Analysis Models (4 Diagrams)	0.0.9	Emre
Appendix D&C , Other Requirements	24/03/26	Added localization, legal requirements, issues list and meetings list.	0.1.0	Beyza
Changed Payment Styles from 2.1, 2.2 and 3.5.2.6	07/04/26	Changed Payment Styles from only In-Store Purchase to In-Store Purchase & In-App Purchase + “EcoBite” to “Süprüz”	0.1.1	Emre
Changed the KM radius 4.1.1	22/04/26	Changed the KM radius in 4.1.1 from 1 to 5	0.1.2	Umut
External Admin Panel 2.3.3	23/04/26	Specified Admin panel to be an external dashboard.	0.1.2.5	Emre
FR-6.2, 4.2.3, 3.5.1, 3.6, 1.4 Has been changed	23/04/26	FR-6.2 Specified 20 KM notification radius, issue 3. solved 4.2.3 Changed offline behaviour to no critical action and only past actions, issue 4 solved 3.5.1 Payment type from physical to both online and physical 3.6 Apple notification and FCM to be removed and only Expo Go will handle both 1.4 Only online payment to both with POS	0.1.3	Emre

Tech Stack & Auth Overhaul	24/03/26	Updated core tech stack (Spring Boot, React Expo, Render, iyzico, Google Maps API). Auth strictly transitioned to Google/Apple SSO: -FR-1.1, FR-1.4,FR-1.6, and NFR-15 deleted. -FR-1.2 (SSO only) and FR-6.7 (FCM only) modified. -Updated text in section 2.4, 3.1.1, 3.6.1, 4.3.1, 4.3.2, and 5.3.1.	0.1.4	Enes
User Documentation & QR Flow Update	27/03/26	Added 2.6.4 Business Partner Onboarding Guide, 2.6.5 No-Show & Cancellation Policy Summary, and 2.6.6 QR Code User Guide. Updated 2.6.1 Consumer Onboarding to reflect dual payment options. Revised 2.6.3 Merchant Success Guidelines to focus on operational best practices, removing overlap with 2.6.4 and 2.6.6. Updated 3.5.2 to distinguish 'Verified / Paid' and 'Verified / Pending Payment' statuses. Added FR-5.9 to 3.5.3. Added NFR-31b and NFR-32b to 5.5.1. Issue 1 is updated and marked as partially resolved.	0.1.5	Beyza

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of Süprüz, a mobile-based platform designed to reduce food waste by enabling businesses to sell unsold yet fresh food items as discounted "surprise packages" near the end of their business hours.

The purpose of this document is to provide a detailed description of the system requirements for **Süprüz Version 1.0**. It serves as a reference for developers, designers, testers, and stakeholders involved in the design, development, deployment, and maintenance of the system.

This SRS defines the expected behavior, system constraints, and user interactions of the Süprüz platform. The document focuses primarily on the mobile application and backend service components that support interactions between customers and participating food businesses. Administrative tools and extended analytics features may be specified in future versions of the system.

1.2 Document Conventions

This document follows common Software Requirements Specification conventions based on **IEEE-style documentation standards**.

The following conventions are used throughout this document:

- **Bold text** is used to highlight important terms, system components, or key concepts.
- *Italic text* is used to introduce new terms or emphasize specific concepts.
- Requirements are written using the keyword **"shall"** to indicate mandatory system behavior.
- Optional or recommended behaviors may use **"should"**.
- System features are organized using numbered headings to ensure traceability.
- Functional requirements will be uniquely identified with an ID format such as **FR-X**, while non-functional requirements will use **NFR-X**.
- Diagrams, tables, and structured lists may be used where necessary to clarify complex interactions.

Unless otherwise specified, **each requirement statement is considered independently prioritized**, allowing future project management tools to assign priorities during planning phases.

1.3 Intended Audience and Reading Suggestions

This document is intended for multiple stakeholders involved in the development and operation of the SÜPRÜZ system, including:

- **Software Developers and Engineers** – who will design and implement the system architecture, backend services, and mobile applications.
- **Project Managers** – who will use the document to plan development phases, allocate resources, and track progress.
- **UI/UX Designers** – who will design the user interface and ensure usability requirements are met.
- **Quality Assurance (QA) Engineers and Testers** – who will derive test cases and validation procedures from the requirements.
- **Business Stakeholders and Product Owners** – who will evaluate whether the system meets business objectives and user needs.
- **Documentation Writers** – who will prepare user manuals and technical documentation.

The structure of this document is organized as follows:

- **Introduction** – provides context, scope, and references for the project.
- **Overall Description** – describes the system environment, users, assumptions, and high-level features.
- **System Features and Functional Requirements** – details the expected system behavior and user interactions.
- **Non-Functional Requirements** – describes performance, security, usability, and reliability requirements.
- **Supporting Information and Appendices** – includes additional diagrams, definitions, or supplementary material.

Readers are recommended to first review the **Project Scope** section to understand the overall objective of the system before proceeding to the detailed requirements.

1.4 Project Scope

Süprüz is a digital platform designed to reduce food waste by connecting food businesses with consumers who are willing to purchase unsold but still fresh food products at significantly discounted prices.

Many restaurants, bakeries, cafes, and grocery stores dispose of unsold food at the end of the day despite the products remaining safe and consumable. Süprüz addresses this issue by enabling businesses to package these items into **discounted "Surprise Packages"** and list them in the application shortly before closing time.

Customers can use the mobile application to:

- Discover nearby businesses offering discounted packages
- View available offers on a map or list
- Purchase packages directly through the platform or from POS (Point Of Sale)
- Pick up their package during a designated time window using a **QR code verification system**

The platform provides several benefits:

- **Consumers**, especially students and budget-conscious users, gain access to affordable meals.
- **Businesses** reduce losses from unsold inventory and attract new customers.
- **Society and the environment** benefit from a reduction in food waste and improved sustainability practices.

The long-term goal of Süprüz is to build a scalable food waste reduction network that supports local businesses while promoting environmentally responsible consumption.

1.5 References

1. **IEEE Computer Society.**
   IEEE Recommended Practice for Software Requirements Specifications (IEEE 830-1998).
   IEEE Standards Association.
2. **Too Good To Go.**
   Official Website.
   https://www.toogoodtogo.com
   (Referenced as an industry example of surplus food marketplace platforms.)
3. **QR Code Specification – ISO/IEC 18004:2015**
   International Organization for Standardization.
   Used as a reference for implementing QR-based package verification.
4. **Sustainability and Food Waste Reports**
   Food and Agriculture Organization (FAO).
   Global Food Losses and Food Waste Report.

2. Overall Description

2.1 Product Perspective

Süprüz is a new, self-contained commercial platform developed to capture an emerging market opportunity in sustainable food tech. While it operates as an independent ecosystem, its Version 1.0 architecture is designed to be highly modular and scalable.

To provide a frictionless transaction experience and establish a scalable commission-based revenue model, Süprüz utilizes a secure in-app online payment system. The system relies on a distributed architecture consisting of a consumer-facing mobile application, a business-facing portal, and a centralized cloud-based backend. Süprüz integrates with key third-party services to function:

- ***Payment Gateways:** Integrates with **iyzico**, a secure third-party payment processor, to handle all in-app transactions and commission disbursements. Sensitive financial data is not stored on Süprüz servers.*
- ***Location and Geofencing Services:** Utilizes mapping APIs (Google Maps API) for precise geolocation, spatial search, and enforcing proximity-based business rules.*
- ***Notification Services:** Leverages solely **Firebase Cloud Messaging (FCM)** to drive user retention through real-time alerts.*

2.2 Product Features

The Süprüz platform provides several high-level functionalities designed to create a frictionless marketplace for surplus food. Detailed specifications are in Section 3.

Major features include:

- ***User Registration and Profile Management:** Frictionless onboarding via **Google and Apple Login** (Single Sign-On) for consumers and business partners. Traditional email/password management is no longer supported.*
- ***Geolocation-Based Discovery:** A dynamic map and list view allowing consumers to locate nearby "Surprise Packages" in real-time.*
- ***Proximity-Based Reservation Engine:** A core USP feature where users actively located within a 5 KM radius of a business can exclusively reserve one "Surprise Package." This drives immediate local foot traffic and prevents no-shows from distant users.*
- ***Inventory and Package Management:** A streamlined dashboard for business staff to instantly create, edit, and publish packages based on daily surplus.*
- ***QR-Based Reservation Verification & In-Store Conversion:** The customer completes the payment via the store's existing Point of Sale (POS) system or the In-App Payment system powered by **iyzico**.*
- ***Push Notifications:** Real-time alerts to notify consumers when local stores list new packages.*

2.3 User Classes and Characteristics

The system is built around three primary user classes, optimized for their specific environments and technical proficiencies:

1. ***Consumers (Target Market / Favored User Class):*** *Mobile users driven by budget-consciousness and sustainability. They require a highly intuitive, location-aware UI, as reserving limited "Surprise Packages" requires them to be physically close to the venue.*
2. ***Business Staff/Managers (Supply Side):*** *Restaurant, cafe, and grocery employees. Operating in fast-paced retail environments, their interface must be ultra-efficient. Verifying reservations via QR scan must seamlessly integrate into their standard physical checkout workflow.*
3. ***System Administrators (Internal Operations):*** *The Süprüz founding and operations team. This class requires an external administrative dashboard to onboard new business partners, monitor system health, and track reservation fulfillment rates.*

2.4 Operating Environment

The Süprüz platform relies on a modern, cloud-based technology stack optimized for rapid development and cross-platform compatibility during its MVP phase:

- ***Frontend Client:*** *Developed using* ***React Native with Expo****, providing a unified and seamless mobile application experience for both iOS and Android platforms from a single codebase.*
- ***Backend Architecture:*** *The core server architecture is built on* ***Spring Boot****, ensuring a robust, scalable environment for handling business logic, transaction management, and API endpoints.*
- ***Database Systems:*** *A centralized* ***PostgreSQL*** *relational database. It crucially utilizes spatial extensions (PostGIS) to execute the high-performance geolocation queries necessary for the 5 KM proximity reservation rule.*
- ***Hosting & Infrastructure:*** *The backend services and database are hosted on* ***Render*** *for streamlined deployment, continuous integration, and operational efficiency.*
- ***Authentication & Identity:*** *The system implements stateless authentication. It relies exclusively on* ***Google and Apple Login*** *(OAuth 2.0). No custom user passwords or primary credential data are stored on Süprüz servers.*
- ***Third-Party Services:***
  - ***Mapping & Location: Google Maps API*** *is integrated for all geocoding, route mapping, and distance calculations.*
  - ***Payments: iyzico*** *serves as the exclusive payment gateway for securely handling in-app transactions.*
  - ***Push Notifications: Firebase Cloud Messaging (FCM)*** *acts as the sole provider for dispatching real-time alerts and notifications across all mobile clients.*

2.5 Design and Implementation Constraints

The MVP development and operation of Süprüz are subject to the following strategic and technical constraints:

- ***Architectural Resource Constraints:*** *As a Version 1.0 product developed by a core 5-person engineering team, the system is strictly constrained to a Modular Monolith architecture. Transitioning to microservices is deferred to future funding and scaling phases to prevent unnecessary operational complexity and infrastructure costs.*
- ***Concurrency and "Flash Reservation" Constraints:*** *The platform must flawlessly handle high-concurrency spikes when popular businesses list "Surprise Packages." Strict database-level locking mechanisms (e.g., pessimistic locking) must be implemented to prevent race conditions and ensure a package is not double-reserved.*
- ***GPS Accuracy and Geofencing Limitations:*** *The core 5 KM reservation rule is inherently constrained by the accuracy of the user's mobile device GPS hardware and signal strength. The system logic must account for potential GPS drift and enforce reasonable location validation to prevent spoofing.*
- ***Hardware Limitations (Business Side):*** *The QR scanning module used by business staff to verify reservations must be heavily optimized to function rapidly and reliably on older mobile devices with lower-quality cameras, particularly in challenging indoor lighting conditions.*

2.6 User Documentation

To ensure seamless adoption and reduce customer support overhead, the following will be provided:

2.6.1 Consumer Onboarding: A frictionless, interactive UI walkthrough presented upon the first launch, clearly explaining the reservation process and available payment options, including both in-app payment and in-store POS payment at the business location.

2.6.2 In-App Help Center: A searchable database of FAQs, focusing on reservation time limits and no-show policies.

2.6.3 Merchant Success Guidelines: A digital handbook for business partners focused on day-to-day operational best practices within the platform. This includes guidance on presenting and packaging surplus food items attractively, setting competitive package prices while meeting the minimum discount requirement, and maximizing consumer engagement with their listings. Registration, onboarding, and QR verification procedures are covered separately in sections 2.6.4 and 2.6.6 respectively.

2.6.4 Business Partner Onboarding Guide :A dedicated onboarding guide shall be provided to assist food businesses through the full process of joining and operating on the Süprüz platform. This guide covers the pre-application phase, the registration and approval process, and the post-approval operational setup.

2.6.4.1 Pre-Application Information Before submitting a registration request, prospective business partners shall be able to access a publicly available information page within the application detailing the general operational model of Süprüz. This page shall explain the overall workflow a business will follow once onboarded and the platform expectations. It shall also outline the general commission structure applicable to business partners, so that applicants are aware of the financial conditions before initiating a formal request. Detailed commission rates and payment mechanisms are specified in a separate financial agreement document provided during the onboarding process.

2.6.4.2 Registration Request Submission Businesses interested in joining the platform shall submit a registration request through the application. The following information shall be collected during this process:

- **Business category and product scope:** The applicant shall specify the category under which their products fall (e.g., bakery, restaurant, grocery) and describe the types of products they can offer within the platform's surprise package model.
- **Surplus frequency:** The applicant shall indicate how frequently unsold food suitable for the platform is generated within their operations, providing a realistic expectation of package availability.
- **Operating hours:** The business's daily working hours shall be submitted, as these directly govern the pickup window constraints enforced by the system (per NFR-33, NFR-35).
- **Location and address:** The physical address of the business shall be provided for geolocation-based discovery and proximity enforcement (per NFR-36).
- **Menu or product range information:** The applicant shall provide menu information or a general description of the food types that may appear in their packages. This requirement is mandatory. Consumers must be able to form a general expectation of the product type they may receive — for example, whether a package may contain baked goods, beverages, or meals — even if the exact contents are not predetermined. Furthermore, the uploaded menu information serves as a reference point for the platform to verify that the listed package price reflects a genuine minimum 50% discount off the original retail value (per NFR-34), thereby protecting both consumer and business rights in the event of a dispute. Businesses may choose to upload their full menu or only the subset of items relevant to surplus packages.
- **External platform presence:** Any active third-party platforms or social media accounts associated with the business shall be provided to assist in identity verification.
- **Contact information:** A valid phone number shall be submitted to enable direct communication during the review process.
- **Official business documentation:** A formal document verifying the legal business registration status of the applicant shall be required as part of the submission.

2.6.4.3 Review and Approval Process Upon submission, the registration request shall be reviewed by Süprüz system administrators. The evaluation shall consider the applicant's product suitability, surplus frequency, operating hours, location, and the completeness and validity of submitted documentation.

If the submitted documents are found to be incomplete or insufficient, administrators shall contact the applicant directly using the provided contact information to request clarification or additional materials. An initial response to the applicant shall be provided within 5 business days of submission. The total duration of the review process may vary depending on the completeness of the submission and the volume of pending applications.

Upon a decision being reached, the applicant shall be notified via the communication channel provided during registration (phone or email) as well as through an in-application notification.

- **If approved:** The business account is activated and the partner may proceed with platform setup.
- **If rejected:** The registration request is considered invalid and the applicant is informed of the outcome. The business may reapply after a mandatory waiting period of one (1) month from the date of rejection.

2.6.4.4 Post-Approval Setup and Business Portal Usage Once approved, business partners shall receive guidance on configuring their presence on the platform and managing day-to-day operations. This includes:

- **Business profile setup:** Partners may customize their business profile page, including display information visible to consumers.
- **Menu and product information management:** Partners shall upload or update their menu or product range. This information is mandatory for consumer-facing visibility and may be updated at any time.
- **Package creation and management:** Partners shall be guided on how to create surprise packages, including setting the original price, discounted price (minimum 50% discount per NFR-34), available quantity, and pickup time window (minimum 30 minutes, within operating hours per NFR-33, NFR-35). Packages may be added, edited, cancelled, or relisted.
- **Reservation and package statistics:** Partners may view their past packages, reservation counts, fulfillment rates, and related operational data through their dashboard.

2.6.5 No-Show & Cancellation Policy Summary: A clear and accessible summary of the platform's no-show and cancellation policies shall be made available to consumers within the application. This summary ensures that users are fully informed of their responsibilities and the consequences of failing to fulfill a reservation.

2.6.5.1 Cancellation Policy: A consumer may cancel an active reservation without penalty provided the cancellation is submitted at least 60 minutes before the start of the pickup window (per NFR-32). Cancellations shall be performed exclusively through the application via the cancellation option on the active reservation screen. Upon a valid cancellation within this time frame, a full refund shall be automatically processed through the platform's payment provider. The refund is expected to be reflected within 3 business days of the cancellation. No additional action from the business partner or system administrators is required for the refund to be initiated.

If the consumer does not cancel within the eligible time window and fails to collect the package, no refund shall be issued and the event shall be recorded as a no-show.

2.6.5.2 No-Show Definition and Detection: A no-show is recorded when a consumer holds an active reservation but does not present their QR code for verification within the designated pickup window (per FR-3.6). The system detects this automatically once the pickup window elapses without a successful QR scan.

Additionally, if a business partner marks a reservation as uncollected — specifying that the consumer did not arrive within the pickup window — this shall also be recorded as a no-show, provided the QR code was never scanned. A business partner may only initiate this action after the pickup window has elapsed. A business partner cannot cancel a reservation after payment has been confirmed and before the pickup window expires; during this period, the reservation remains active and the package is held for the consumer.

If a business partner publishes an incorrect package listing and cancels it before any reservations are made, this shall not affect any consumer's no-show record.

2.6.5.3 No-Show Consequences and Suspension No-show events are tracked on a rolling 30-day basis. The following rules apply:

- Upon accumulating **3 no-shows within a 30-day period**, the consumer's reservation privileges shall be automatically suspended for **7 days** (per NFR-31, FR-3.7).
- Upon accumulating a **second suspension** within the platform's history for that account, the suspension shall become **permanent**.
- Once a 7-day suspension period expires, the consumer's reservation privileges shall be automatically restored and the user shall be notified via a push notification (per FR-6.6).
- The no-show counter resets automatically after the 30-day rolling period. However, the full no-show and reservation history remains visible to system administrators.

2.6.5.4 Suspension Experience During an active suspension period, the consumer shall not be able to access or use the application. Any login attempt made during this period shall display a message informing the user of their suspended status and the remaining time until the suspension is lifted. Discovery and browsing of available packages shall not be accessible during suspension.

2.6.5.5 Reservation History and Transparency: While the application does not display an explicit no-show counter to the consumer, users may review their full reservation history from their profile screen. This history includes the status of each past reservation, categorized as completed, cancelled, or no-show, allowing consumers to track their own record (per FR-1.5).

2.6.6 QR Code User Guide (Consumer-Facing) This guide provides consumers with a practical overview of how the QR code system works within the Süprüz application, covering the full lifecycle from reservation to package pickup.

2.6.6.1 QR Code Generation A unique QR code is automatically generated and assigned to the consumer immediately upon a successful reservation. The QR code is accessible at any time from the active reservation screen within the application and remains visible throughout the pickup window. The QR code is unique to the specific reservation and the specific business, meaning it cannot be used at any other location or for any other package.

2.6.6.2 Validity and Expiry The QR code is valid exclusively within the pickup time window defined by the business partner at the time of listing. A live countdown timer displayed on the reservation screen shows the remaining time before the QR code expires. Once the pickup window elapses, the QR code becomes invalid and unscannable. Consumers are advised to arrive within the designated pickup window to avoid automatic cancellation of their reservation and the recording of a no-show event (per Section 2.6.5).

2.6.6.3 Payment and QR Code Relationship The QR code serves as proof of reservation and is independent of the payment method chosen by the consumer. Two payment scenarios are supported:

- **In-app payment:** If the consumer completes payment through the application prior to arriving at the business, the QR code will reflect a 'Verified / Paid' status upon scanning. The business partner will hand over the package without requiring further payment.
- **In-store POS payment:** If the consumer opts to pay at the business location, the QR code will reflect a 'Verified / Pending Payment' status upon scanning. The business partner will then process the payment via their Point-of-Sale system before handing over the package.

In both cases, the consumer is only required to open the reservation screen and present the QR code to the business staff for scanning.

2.6.6.4 Requirements for Successful Scanning For the QR code to be scanned successfully, the following conditions must be met:

- **Active internet connection:** The application requires an active internet connection at all times. The QR code screen cannot be accessed in offline mode. Consumers should ensure they have a stable connection before arriving at the business location.

- **Screen visibility:** Consumers should ensure their device screen is clean and visible. If scanning difficulty is encountered, increasing the screen brightness manually may assist the business staff in completing the scan.
- **Single use:** Each QR code can only be scanned once. Once a successful scan has been recorded, the reservation is marked as fulfilled and the QR code becomes permanently inactive.

2.6.6.5 What To Do If Something Goes Wrong If a consumer encounters an issue during the QR verification process, the following guidance applies:

- If the QR code appears expired but the consumer believes they arrived within the pickup window, they should contact Süprüz support through the in-app Help Center.
- If the business staff reports a scanning failure, the consumer should ensure their screen brightness is adequate and that the QR code is fully visible on screen.
- Technical failures during QR scanning that are outside the consumer's control may be reviewed by system administrators. Consumers should retain their reservation confirmation notification as a reference.

2.7 Assumptions and Dependencies

The successful launch and operation of Süprüz rely on the following:

- ***Assumptions:*** *Consumers have access to internet-connected smart devices with active and accurate GPS.*
    - *Partner businesses maintain reliable internet access and at least one smart device on-premises for QR verification.*
    - *Users will fulfill their reservations; a penalty system for "no-shows" will be necessary to maintain business trust.*
- ***Dependencies:***
    - ***Third-Party APIs:*** *System functionality is highly dependent on the uptime and accuracy of the mapping/geolocation provider.*
    - ***Mobile OS Ecosystems:*** *Deployment, updates, and user acquisition are subject to the review processes and policies of the Apple App Store and Google Play Store, particularly regarding location permission justifications.*

3. System Features

This section details the functional requirements of the Süprüz platform, organized by system feature. Each feature includes a description with assigned priority, stimulus/response sequences illustrating system behavior, and a table of traceable functional requirements.

3.1 User Registration and Profile Management

3.1.1 Description and Priority

This feature enables new users (both Consumers and Business Partners) to create an account and manage their profile on the Süprüz platform. It supports Single Sign-On (SSO) via Google or Apple accounts for a frictionless onboarding experience, as well as traditional email/password registration. Users can update personal details, manage notification preferences, and view their reservation history from their profile.

Priority	High
Benefit	9 — Core gateway feature; no user activity is possible without an account.
Penalty	9 — System is entirely unusable without registration.
Cost	4 — SSO integration requires third-party OAuth setup but is well-documented.
Risk	3 — Low risk; mature authentication libraries available.

3.1.2 Stimulus/Response Sequences

#	Actor	Action / System Response
1	Consumer / Business Staff	Opens the Süprüz application for the first time.
2	System	Displays the Welcome Screen with options: 'Continue with Google', 'Continue with Apple', and 'Sign Up with Email'.
3	User	Selects an SSO provider (e.g., Google).
4	System	Redirects the user to the OAuth 2.0 authorization page of the selected provider.
5	User	Grants permission on the provider's page.
6	System	Receives the authorization token, creates a new user record, and redirects the user to the Role Selection screen (Consumer or Business Partner).
7	User	Selects their role.
8	System	Creates the user profile with the assigned role and presents the onboarding walkthrough.
E1	System	If the OAuth token is invalid or expired, displays an error message: 'Authentication failed. Please try again.'

E2	System	If the email is already registered via a different method, prompts the user to log in using their existing method.

3.1.3 Functional Requirements

Req. ID	Requirement Description	Priority
FR-1.1	The system shall support Single Sign-On (SSO) registration and login via Google and Apple OAuth 2.0.	High
FR-1.2	The system shall require users to select a role (Consumer or Business Partner) during the initial registration flow.	High
FR-1.3	The system shall allow authenticated users to update their display name, profile picture, and notification preferences.	Medium
FR-1.4	The system shall allow users to permanently delete their account and associated personal data in compliance with data privacy regulations.	Medium
FR-1.5	The system shall display a user's reservation history, including past and upcoming packages, within their profile screen.	Medium

3.2 Geolocation-Based Discovery

3.2.1 Description and Priority

This feature allows Consumer users to discover nearby businesses offering Surprise Packages in real time. The system retrieves the user's current GPS location and queries the backend for active packages within a configurable radius. Results are displayed both on an interactive map view (with business pins) and as a sortable list view. The feature is central to Süprüz's value proposition of connecting local supply with local demand.

Priority	High
Benefit	9 — Primary discovery mechanism; drives all consumer-side engagement.
Penalty	8 — Without discovery, consumers cannot access any packages.
Cost	6 — Requires mapping API integration and spatial database queries (PostGIS).
Risk	5 — Dependent on third-party mapping API availability and device GPS accuracy.

3.2.2 Stimulus/Response Sequences

#	Actor	Action / System Response
1	Consumer	Opens the Süprüz application and navigates to the 'Discover' screen.
2	System	Requests location permission from the device OS if not already granted.
3	Consumer	Grants location permission.
4	System	Retrieves the user's current GPS coordinates and sends a spatial query to the backend.
5	System	Returns a list of businesses with active Surprise Packages within the configured discovery radius (default: 5 KM) and renders them as pins on the map.
6	Consumer	Taps a business pin or list item.
7	System	Displays the business detail card showing: business name, package description, available quantity, pickup time window, and discounted price.
E1	System	If location permission is denied, displays an informational banner: 'Enable location access for the best experience' and defaults to a manual city search.
E2	System	If the mapping API is unavailable (per NFR-13), gracefully degrades to a list-only view with a notice: 'Map is temporarily unavailable.'

3.2.3 Functional Requirements

Req. ID	Requirement Description	Priority
FR-2.1	The system shall request and utilize the device's GPS location to populate the discovery map with nearby businesses.	High
FR-2.2	The system shall display active Surprise Package listings on an interactive map view, with each business represented by a distinct map pin.	High
FR-2.3	The system shall provide an alternative list view displaying the same businesses sorted by distance from the user's current location.	High
FR-2.4	The system shall display a business detail card upon selection, containing: business name, category, package description, available quantity, pickup time window, and price.	High
FR-2.5	The system shall automatically refresh available listings on the discovery map at a maximum interval of 60 seconds without requiring a manual user action.	Medium
FR-2.6	The system shall allow users to filter listings by category (e.g., Bakery, Restaurant, Grocery) and sort by distance or pickup time.	Medium

FR-2.7	The system shall gracefully degrade to a list-only view when the mapping API is unavailable, maintaining core discovery functionality.	High
FR-2.8	The system shall allow users who have denied location permission to manually search for businesses by entering a city or neighborhood.	Low

3.3 Proximity-Based Reservation Engine

3.3.1 Description and Priority

This is the core differentiating feature of Süprüz. It allows a Consumer to reserve a Surprise Package exclusively when they are physically within a 5 KM radius of the listing business. This proximity enforcement drives immediate local foot traffic, reduces no-shows from distant users, and protects the business model. The reservation process must handle high-concurrency scenarios to prevent double-bookings through database-level locking mechanisms.

Priority	High
Benefit	9 — Core business differentiator; directly enables the local commerce model.
Penalty	9 — Without this, the platform becomes a generic listing app, losing its USP.
Cost	7 — Complex concurrency control and geofencing validation logic required.
Risk	7 — GPS spoofing and race conditions are active threat vectors requiring mitigation.

3.3.2 Stimulus/Response Sequences

#	Actor	Action / System Response
1	Consumer	Taps the 'Reserve' button on a business detail card.
2	System	Retrieves the consumer's current GPS coordinates in real time.
3	System	Validates that the consumer's location is within a 5 KM radius of the selected business (server-side geofencing check).
4	System	Acquires a pessimistic lock on the selected package's inventory record in the database.

5	System	Confirms available quantity > 0 and the consumer has not exceeded the daily reservation limit (NFR-30).
6	System	Creates a reservation record, decrements the available package count by 1, and releases the database lock.
7	System	Returns a success confirmation to the consumer and triggers the QR code generation flow (see Feature 3.5).
E1	System	If the consumer is outside the 5 KM radius, displays: 'You must be within 5 KM of this business to reserve a package.'
E2	System	If the package quantity reaches 0 concurrently (race condition resolved), displays: 'Sorry, this package just sold out.'
E3	System	If the daily limit is reached, displays: 'You have reached your daily reservation limit of 2 packages.'

3.3.3 Functional Requirements

Req. ID	Requirement Description	Priority
FR-3.1	The system shall validate the consumer's real-time GPS coordinates server-side before processing any reservation request to enforce the 5 KM proximity rule.	High
FR-3.2	The system shall reject any reservation attempt from a consumer located more than 5 KM from the target business and return a descriptive error message.	High
FR-3.3	The system shall apply a pessimistic database lock on the target package inventory record for the duration of a reservation transaction to prevent double-booking.	High
FR-3.4	The system shall enforce a maximum of two (2) active reservations per Consumer account per calendar day (per NFR-30).	High
FR-3.5	The system shall allow a Consumer to cancel a reservation without penalty provided the cancellation is submitted at least 60 minutes before the pickup window (per NFR-32).	High
FR-3.6	The system shall automatically cancel an unverified reservation if the consumer does not present their QR code within the pickup time window, and shall record this event as a no-show.	High
FR-3.7	The system shall automatically suspend a consumer's reservation privileges for 7 days upon accumulating three (3) no-shows within a 30-day rolling period (per NFR-31).	High
FR-3.8	The system shall display a real-time countdown timer on the active reservation screen showing the remaining time within the pickup window.	Medium

3.4 Inventory and Package Management

3.4.1 Description and Priority

This feature provides Business Staff and Managers with a streamlined dashboard to create, publish, edit, and close Surprise Package listings based on their daily surplus inventory. Given that businesses operate in fast-paced environments near the end of their business hours, the interface is designed for maximum speed and efficiency. The system enforces platform business rules such as the minimum 50% discount and valid pickup windows before a package can be published.

Priority	High
Benefit	9 — Supply-side feature; without it, there are no packages for consumers to discover.
Penalty	9 — A complex or slow interface will discourage business adoption.
Cost	5 — Standard CRUD dashboard with business rule validation.
Risk	3 — Low risk; well-understood CRUD patterns apply.

3.4.2 Stimulus/Response Sequences

#	Actor	Action / System Response
1	Business Staff	Logs into the Business Portal and navigates to 'Manage Packages'.
2	System	Displays the package dashboard showing currently active, upcoming, and past listings.
3	Business Staff	Taps '+ New Package'.
4	System	Presents a package creation form with fields: description, original retail price, quantity, pickup start time, and pickup end time.
5	Business Staff	Fills in the form and taps 'Publish'.
6	System	Validates the input: discounted price >= 50% off, pickup window >= 30 minutes, window within registered operating hours (per NFR-33, NFR-34, NFR-35).
7	System	Saves the package record, makes it visible to consumers on the discovery map, and triggers a push notification dispatch (Feature 3.6).
E1	System	If the discount is below 50%, displays: 'The package price must reflect at least a 50% discount from the original retail price.'
E2	System	If the pickup window is less than 30 minutes, displays: 'Pickup window must be at least 30 minutes.'
E3	System	If the pickup window extends beyond registered operating hours, displays: 'Pickup window cannot extend beyond your registered closing time.'

3.4.3 Functional Requirements

Req. ID	Requirement Description	Priority
FR-4.1	The system shall allow authenticated Business Staff to create a new Surprise Package listing with a description, original price, quantity, and pickup time window.	High
FR-4.2	The system shall reject any package listing where the set price does not reflect a minimum 50% discount off the stated original retail price (per NFR-34).	High
FR-4.3	The system shall reject any package listing with a pickup window of less than 30 minutes in duration (per NFR-33).	High
FR-4.4	The system shall prevent Business Staff from setting a pickup window that extends beyond the business's officially registered operating hours (per NFR-35).	High
FR-4.5	The system shall allow Business Staff to edit the description or quantity of a package listing while it is active, provided no reservations have been made.	Medium
FR-4.6	The system shall allow Business Staff to manually close a package listing at any time, which shall notify all consumers with active reservations for that package.	High
FR-4.7	The system shall display a real-time counter of active reservations and remaining quantity on each listing in the business dashboard.	Medium
FR-4.8	The system shall automatically remove a listing from the public discovery map once the pickup time window has elapsed.	High

3.5 QR-Based Reservation Verification and In-Store Conversion

3.5.1 Description and Priority

Upon a successful reservation, the system generates a unique, encrypted, and time-limited QR code for the Consumer. At the physical business location, Business Staff scan this QR code using the Süprüz Business Portal to verify the reservation. Upon successful verification, the transaction is converted into a completed sale, and the Consumer pays through either the store's existing Point-of-Sale (POS) system or from the App’s Online Payment System. This flow is the critical link between the digital reservation and the physical transaction.

Priority	High
Benefit	9 — Closes the transaction loop; without it, reservations cannot be fulfilled.
Penalty	9 — Failure here directly affects business revenue and consumer trust.
Cost	6 — Encrypted QR generation with time-stamping and camera optimization needed.
Risk	6 — QR spoofing via screenshots; must be mitigated with time-limited encryption.

3.5.2 Stimulus/Response Sequences

#	Actor	Action / System Response
1	Consumer	Arrives at the business and opens their active reservation in the Süprüz app.
2	System	Displays the consumer's unique, encrypted QR code and a countdown to pickup window expiry.
3	Business Staff	Opens the QR Scanner within the Business Portal and scans the consumer's QR code.
4	System	Decrypts and validates the QR code: verifies the timestamp (not expired), reservation ID, and business ID match.
5	System	System Updates the reservation status based on payment method: if the consumer has completed in-app payment prior to arrival, the status is updated to 'Verified / Paid'; if payment is to be collected in-store via POS, the status is updated to 'Verified / Pending Payment'. A success confirmation is displayed to Business Staff in both cases.
6	Business Staff	Hands over the package after the QR Confirmation and processes payment via the store's POS system.
7	System	Upon Business Staff confirming pickup in the portal, updates the reservation status to 'Completed' on both the Consumer and Business dashboards.
E1	System	If the QR code has expired (pickup window has passed), displays: 'This reservation has expired.'
E2	System	If the QR code has already been scanned, displays: 'This reservation has already been fulfilled.'
E3	System	If the QR code belongs to a different business, displays: 'Invalid code. This reservation is not for this location.'

3.5.3 Functional Requirements

Req. ID	Requirement Description	Priority

FR-5.1	The system shall generate a unique, encrypted QR code for each confirmed reservation immediately after the reservation is created.	High
FR-5.2	The generated QR code shall be dynamically time-stamped and cryptographically signed to prevent fraudulent use of intercepted screenshots (per NFR-21).	High
FR-5.3	The QR code shall become invalid and unscannable once the pickup time window has elapsed.	High
FR-5.4	The Business Portal's QR scanner shall verify a reservation within 2 seconds of a successful scan (per NFR-3).	High
FR-5.5	The system shall prevent a QR code from being scanned and accepted more than once (idempotency).	High
FR-5.6	The system shall display a clear success or failure result to the Business Staff immediately following a scan attempt.	High
FR-5.7	The QR scanning module shall be optimized to function reliably on older mobile devices with lower-quality cameras in indoor lighting conditions (per design constraint).	High
FR-5.8	The system shall update the reservation status to 'Completed' on both the Consumer and Business dashboards upon Business Staff confirming the final handover.	Medium
FR-5.9	The system shall distinguish between in-app pre-payment and in-store POS payment at the point of QR verification, updating the reservation status to 'Verified / Paid' or 'Verified / Pending Payment' accordingly.	High

3.6 Push Notifications

3.6.1 Description and Priority

Push notifications are delivered exclusively via **Firebase Cloud Messaging (FCM)** for both Android and iOS devices.

Priority	Medium
Benefit	8 — Key driver of user retention and timely reservation fulfillment.
Penalty	5 — System is functional without it, but user engagement will be significantly lower.
Cost	4 — Expo Go is well-documented and has an existing SDK.
Risk	2 — Low risk; relies on stable and mature third-party notification services.

3.6.2 Stimulus/Response Sequences

#	Actor	Action / System Response
1	Business Staff	Publishes a new Surprise Package listing (see Feature 3.4).
2	System	Queries the backend for all Consumer accounts whose last known location is within a configurable notification radius of the business.
3	System	Dispatches a push notification to all qualifying Consumer devices via Expo Go within 2 seconds of publication (per NFR-6).
4	Consumer	Receives a push notification: 'New package available near you at [Business Name]!'
5	Consumer	Taps the notification.
6	System	Deep-links directly to the relevant business detail card in the Discovery map.
7	System	Sends a reminder push notification 30 minutes before the consumer's active reservation pickup window closes.
E1	System	If a consumer has disabled notifications for Süprüz, the notification is silently dropped; no error is raised.

3.6.3 Functional Requirements

Req. ID	Requirement Description	Priority
FR-6.1	The system shall dispatch a push notification to Consumer devices within 2 seconds of a Partner Business publishing a new Surprise Package (per NFR-6).	High
FR-6.2	Push notifications for new listings shall only be sent to Consumer accounts whose last known location is within a 20 KM notification radius of the publishing business.	High
FR-6.3	The system shall send a reservation confirmation push notification to the Consumer immediately after a successful reservation.	High
FR-6.4	The system shall send a reminder push notification to the Consumer 30 minutes before the pickup window of their active reservation closes.	Medium
FR-6.5	The system shall notify the Consumer immediately if their reservation is cancelled by the business (e.g., the listing is closed).	High
FR-6.6	The system shall notify the Consumer immediately if their reservation privileges are suspended due to accumulated no-shows.	Medium
FR-6.7	The system shall support notification delivery centrally via Firebase Cloud Messaging (FCM) for both Android and iOS platforms.	High
FR-6.8	The system shall allow Consumers to manage their notification preferences (enable/disable specific notification types) from the profile settings screen.	Low

4. External Interface Requirements

4.1 User Interfaces

The Süprüz platform exposes two primary user-facing interfaces—a Consumer Mobile Application and a Business Partner Portal—each tailored to the workflows and technical proficiency of its target user class.

4.1.1 Consumer Mobile Application (iOS & Android)

The Consumer application shall present the following core screens and UI components:

A. Discovery Screen (Main View)
- Description: The primary landing screen after user login where they find active surplus packages.
- Layout / Elements:
  - ➢ Top header carrying a Category filter carousel (Bakery, Restaurant, etc.) and a 'View Toggle' button.
  - ➢ Central interactive map pane spanning the screen.
  - ➢ Geolocation pins to indicate businesses with currently active listings.
  - ➢ Bottom persistent navigation bar with "Discover, Reservations, Profile" tabs.
- Inputs:
  - ➢ Tap/Swiping gestures to pan and zoom the map.
  - ➢ Tapping a category chip to filter results.
  - ➢ Tapping a map pin to inspect a business.
- Outputs / Responses: Tapping the 'View Toggle' switches to a paginated List View of cards. Tapping a map pin instantly highlights the pin and displays the Business Detail Card modal overlay with animation.
- Navigation: Directs the user to the Business Detail Card for specific package reservation.

B. Business Detail Card
- Description: A bottom-sheet overlay presenting specific package details for a selected business.
- Layout / Elements:
  - ➢ Hero image of the business facility.
  - ➢ Informational section: Business name, category, rating, and straight-line distance.

  - ➢ Package section: Brief package description, original retail price crossed out, prominent discounted price (-50% badge per NFR-34).
  - ➢ Status section: Remaining quantity integer and pickup time window.
  - ➢ A conditional dynamic "Reserve Now" Action Button.
  - ➢ A proximity warning banner visible only if the user is > 5 KM away.
- Inputs: Tapping the "Reserve Now" button.
- Outputs / Responses: If successful, the screen presents a visual confirmation dialog. If the user is outside the 5 KM geofence (per NFR-36), the button is disabled and the warning reads "You must be within 5 KM to reserve".
- Navigation: Accessible from Discovery Screen. Leads to the Reservation Screen upon success.

C. Reservation Screen
- Description: The primary verification utility for the consumer to facilitate the in-store transaction.
- Layout / Elements:
  - ➢ A large, high-contrast, dynamically generated QR code centered on the screen (see Feature 3.5).
  - ➢ A critical live countdown timer indicating validity relative to the pickup window limit.
  - ➢ Status badge indicators (e.g., 'Reserved', 'Completed', 'No-Show').
  - ➢ Order summary constraints (Price, Package type).
  - ➢ A "Cancel Reservation" Action Button.
- Inputs: Tapping "Cancel Reservation".
- Outputs / Responses: A prompt checks cancellation rules (minimum 60 minutes before the window per NFR-32). If valid, prompts with "Are you sure?".
- Navigation: Returns to Consumer Dashboard / History if canceled or completed.

D. Profile & Settings Screen
- Description: A dedicated administrative portal for managing consumer identities and behaviors.
- Layout / Elements:
  - ➢ Header module displaying Avatar, Display Name, and Email.
  - ➢ "Eco-Impact" tracker widgets: Metric count for 'Packages Saved', 'Money Saved (₺)', 'Food Saved (kg)'.
  - ➢ Notification toggle switches (Push Notifications, New Package Alerts, Pickup Reminders).
  - ➢ "Delete Account" and "Logout" buttons formatted as critical destructive actions (red coloring).
- Inputs: Editing inputs on text fields, boolean toggles on sliders.
- Outputs / Responses: Notification preferences update immediately upon toggling via API PUT request.
- Navigation: Accessible via the persistent bottom tabs.

E. Onboarding Walkthrough
    - Description: First-contact educational material introducing the platform structure.
    - Layout / Elements:
        - ➢ Full-screen swipe-able paged carousel showing 4 educational graphics.
        - ➢ Paging dots indicator.
        - ➢ "Skip" actionable text and "Get Started" primary action button.
    - Inputs: Horizontal swipe actions, tap on buttons.
    - Outputs / Responses: Transitions between educational visual slides.
    - Navigation: Shown once upon fresh installation; leads permanently to Auth / Registration forms.

4.1.2 Business Partner Portal (Mobile & Web)

The Business Partner interface maintains synchronization across responsive web dashboards and dedicated mobile apps, designed for fast-paced retail environments.

A. Package Management Dashboard
    - Description: The central hub for Business Staff to overview operational status.
    - Layout / Elements:
        - ➢ Global "At-A-Glance" widgets aggregating active listings, total reservations, and daily revenue.
        - ➢ Scrollable list of Active Package Cards, each possessing inventory counters ("3/5 Reserved") and active countdowns.
        - ➢ Distinct "+ New Package" primary action floating button.
        - ➢ Card-specific buttons: "Edit", "Close Listing".
    - Inputs: Tapping action buttons.
    - Outputs / Responses: Inventory quantities are pulled in real-time, refreshing visually every 30 seconds.
    - Navigation: Links staff directly to Package Creation or Reservation Management screens.

B. Package Creation Form
    - Description: Data entry UI for rapidly submitting food surplus into the marketplace.
    - Layout / Elements:
        - ➢ Form fields for: Package Description (Multiline), Original Retail Price (₺), Discounted Price (₺), and Stock Quantity.
        - ➢ Graphical Time Pickers for the 'Pickup Start' and 'Pickup End' margins.
        - ➢ Inline dynamic Discount Indicator (e.g., "70% discount ✓").
        - ➢ A "Publish Package" submission action.
    - Inputs: Keyboard numeric/alphanumeric strings, modal time-picker interactions.
    - Outputs / Responses: The system calculates the discount percentage concurrently as characters are typed. If the discount < 50% or window < 30 minutes, field borders turn red with appropriate inline warnings blocking submission.
    - Navigation: Accessible from Dashboard. Returns to Dashboard upon successful backend POST.

C. QR Scanner Interface
    - Description: The pivotal transaction screen enabling business staff to verify consumer ownership upon arrival.
    - Layout / Elements:
        - ➢ A full-screen live camera viewfinder feed taking up 80% of lateral geometry.
        - ➢ A superimposed semi-transparent target framework "Point camera at QR code".
        - ➢ A dynamic state overlay covering the entire screen when a scan returns.
    - Inputs: Activating the device's rear-facing objective lens over consumer screens.
    - Outputs / Responses:
        - ➢ Success Event: Within 2 seconds (per NFR-3), the viewfinder transforms into an opaque green overlay exhibiting a large checkmark icon and rendering the customer's name and expected payment amount.
        - ➢ Failure Event: Displays an opaque red interface reporting "Expired" or "Invalid Code" alongside a vibrating haptic device feedback pattern.
    - Navigation: Accessible directly from the bottom navigation bar. Resets to scanning state upon dismissal.

D. Reservation Management View
    - Description: A tabular accounting ledger exposing all real-time reservations.
    - Layout / Elements:
        - ➢ Paginated or infinite-scrolling list entries.
        - ➢ Data points per entry: Consumer Display Name, System Reservation ID (#XXXX), Active Status Pill (Blue for 'Reserved', Green for 'Completed', Red for 'No-Show', Grey for 'Cancelled').
        - ➢ Financial impact integer (₺).
    - Inputs: Sorting headers or filter chips.
    - Outputs / Responses: Automatically shifts an item's status pill from Reserved -> Completed as the backend acknowledges successful QR scans.
    - Navigation: Serves as an analytical endpoint without deep sequential progression.

4.1.3 General User Interface Standards

- All interfaces shall conform to platform-specific Human Interface Guidelines (Apple HIG for iOS, Material Design for Android).
- Touch targets shall be no smaller than 44×44 points (iOS) / 48×48 dp (Android) to support single-handed, fast-paced operation in retail environments.
- Interfaces shall scale responsively across varying device resolutions and shall support both light and dark display modes.
- Font sizes, color contrast ratios, and interactive element spacing shall comply with WCAG 2.1 Level AA accessibility standards (per NFR-26).
- The Consumer application shall allow a user to complete a reservation in no more than 3 screen taps from the Discovery map (per NFR-24).

4.2 Hardware Interfaces

Süprüz does not interface with proprietary or custom hardware. Instead, it relies on standard hardware subsystems available on modern commercial mobile devices. The following hardware components are accessed through operating-system-provided APIs:

4.2.1 GPS / Location Services Module

- Purpose: Determines the consumer's real-time geographic coordinates for the 5 KM proximity-based reservation enforcement (per FR-3.1, NFR-36) and for populating the discovery map with nearby listings.
- Interface Method: The application shall access the device GPS through the OS-native location services API (CoreLocation on iOS, Fused Location Provider on Android).
- Constraints: The system shall use OS-native geofencing APIs rather than continuous GPS polling to minimize battery consumption (per NFR-9). Server-side validation shall account for potential GPS drift of up to ±50 meters.

4.2.2 Camera Module

- Purpose: Enables Business Staff to scan Consumer QR codes during the in-store verification and pickup process (per Feature 3.5).
- Interface Method: The Business Portal shall access the device's rear-facing camera through the OS camera API (AVFoundation on iOS, CameraX on Android) or the browser MediaDevices API in the web portal.
- Constraints: The QR scanning module shall be optimized to function reliably on devices with lower-quality cameras (minimum 5 MP) and in challenging indoor lighting conditions with low ambient light (per FR-5.7).

4.2.3 Network Interface (Cellular / Wi-Fi)

- Purpose**:** Provides the data connectivity required for all client-server communication, including real-time reservation transactions, map data retrieval, and push notification delivery.
- Interface Method**:** Standard OS-level TCP/IP networking stack over cellular (4G LTE / 5G) or Wi-Fi adapters.
- Constraints**:** An active internet connection is mandatory for all core operations, including reservation creation, verification, and discovery services. When connectivity is lost, the application shall display a clear offline indicator and switch to read-only offline mode. In offline mode, users may view previously loaded data such as past reservations, saved profiles, and recent activity, but creation, modification, or submission of new actions shall be disabled until connectivity is restored.

4.3 Software Interfaces

Süprüz integrates with several external software services and libraries. This section specifies each dependency, the interface contract, and version requirements.

4.3.1 Mapping & Geolocation API

- Software: Google Maps SDK (as referenced in Section 2.1).
- Purpose: Renders the interactive discovery map, provides geocoding services, calculates straight-line distances between consumer and business coordinates, and supplies the visual pin and route overlays.
- Data Exchanged: The client application sends the user's latitude/longitude coordinates and receives map tile data, business marker positions, and distance calculations in return.
- Interface: REST API calls over HTTPS and native mobile SDK method calls.
- Fallback: If the mapping API is unavailable, the system shall gracefully degrade to a list-only discovery view (per NFR-13, FR-2.7).

4.3.2 Push Notification Services

- Software: Firebase Cloud Messaging (FCM) for both Android and iOS devices.
- Purpose: Delivers real-time server-triggered notifications to Consumer devices for events including new package availability, reservation confirmation, pickup reminders, and cancellation alerts (per Feature 3.6).
- Data Exchanged: The backend dispatches notification payloads containing a title, body message, deep-link URI, and optional metadata (e.g., business ID, reservation ID) to FCM servers, which relay them to registered device tokens.
- Interface: FCM HTTP v1 API over TLS-secured connections.
- Constraint: Notifications shall be dispatched within 2 seconds of the triggering event (per NFR-6).

4.3.3 OAuth 2.0 Identity Providers

- Software: Google Identity Services; Apple Sign In.
- Purpose: Provides federated Single Sign-On (SSO) authentication, allowing users to register and log in without creating a separate password (per FR-1.2).
- Data Exchanged: The client application initiates an OAuth 2.0 authorization code flow. Upon user consent, the identity provider returns an authorization code which the backend exchanges for an ID token containing the user's verified email address, display name, and a unique subject identifier.
- Interface: OAuth 2.0 / OpenID Connect protocol endpoints over HTTPS.
- Constraint: If the OAuth token is invalid or expired, the system shall display an authentication error and prompt the user to retry (per Stimulus/Response E1 in Section 3.1.2).

4.3.4 Relational Database with Spatial Extensions

- Software: PostgreSQL (version 14+) with the PostGIS spatial extension.
- Purpose: Serves as the primary persistent data store for all application data including user accounts, business profiles, package listings, and reservation records. The PostGIS extension enables efficient spatial queries required for the 5 KM proximity logic (per NFR-36) and the radius-based discovery feature (per Feature 3.2).
- Data Exchanged: The backend application server communicates with the database using parameterized SQL queries over a persistent connection pool (per NFR-7). Spatial queries use PostGIS functions (e.g., ST_DWithin) to resolve distance-based lookups.
- Interface: PostgreSQL wire protocol (libpq) over a secured internal network connection.
- Constraint: Spatial queries shall return results in under 500 milliseconds (per NFR-1). The connection pool shall sustain 10,000 concurrent read operations during peak hours (per NFR-7).

4.3.5 Containerization Platform

- Software: Docker Engine.
- Purpose: Packages the backend modular monolith and its dependencies into portable containers, ensuring consistent behavior across development, testing, and production environments (per NFR-27, Section 2.4).
- Interface: Dockerfiles define the build specification; Docker Compose or equivalent orchestration manages multi-container deployments.

4.4 Communications Interfaces

This section defines the communication protocols, data formats, and transport-level security standards governing all data exchange within and around the Süprüz platform.

4.4.1 Transport Protocol & Encryption

- All communication between mobile clients, the business web portal, and the cloud backend shall occur exclusively over HTTPS, encrypted using Transport Layer Security (TLS) version 1.2 or higher (per NFR-20).
- Plaintext HTTP connections shall be explicitly rejected by the backend; HTTP Strict Transport Security (HSTS) headers shall be enforced.

4.4.2 API Architecture & Data Format

- The backend shall expose a RESTful API adhering to standard HTTP semantics (GET, POST, PUT, DELETE) with resource-oriented URL structures.

- All request and response payloads shall be formatted in JavaScript Object Notation (JSON) with UTF-8 encoding.
- The API shall return appropriate HTTP status codes (e.g., 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 409 Conflict, 429 Too Many Requests) to communicate the outcome of each request.
- API versioning shall be implemented via URL path prefixes (e.g., /api/v1/) to support backward-compatible evolution of the interface.

4.4.3 Push Notification Delivery

- Outbound push notifications shall be dispatched from the backend to FCM and APNs gateway servers over their respective HTTPS APIs (as specified in Section 4.3.2).
- Notification payloads shall conform to the FCM Message format and the APNs payload structure, including the required fields: notification title, body, and an optional data payload containing a deep-link URI.

4.4.4 Rate Limiting & Throttling

- The API gateway shall enforce rate limiting of a maximum of 60 requests per minute per client IP address (per NFR-18) to protect against brute-force attacks and denial-of-service attempts.
- Clients exceeding the rate limit shall receive an HTTP 429 (Too Many Requests)response with a Retry-After header indicating when requests may resume.

4.4.5 Session & Token Management

- Authenticated sessions shall be managed using JSON Web Tokens (JWT) transmitted in the HTTP Authorization header using the Bearer scheme.
- Access tokens shall have a maximum validity of 24 hours (per NFR-16). Refresh tokens shall be stored securely on the device using platform-specific secure storage (Keychain on iOS, EncryptedSharedPreferences on Android).
- All active sessions shall be invalidated immediately upon a user-initiated password reset (per NFR-17).

5. Other Nonfunctional Requirements

5.1 Performance Requirements

*The performance requirements of **Süprüz** define the expected system responsiveness, throughput, capacity, and resource efficiency to ensure a seamless experience during time-sensitive Surprise Package reservations.*

5.1.1 Response Time & Latency

- **NFR-1:** The **Database** *shall* process spatial queries required to display available *Surprise Packages* within the 5 KM radius and return results to the **Client Application** in under 500 milliseconds.
- **NFR-2:** Standard REST API requests between the **Mobile Application** and the **Cloud Backend** *should* achieve a response time of under 300 milliseconds for 95% of normal load requests.
- **NFR-3:** The **QR Scanning Module** *shall* successfully verify reservations within 2 seconds to ensure uninterrupted physical checkout workflows.
- **NFR-4:** The **Mobile Application** *shall* render the primary map interface and fully populate local pins within 1.5 seconds of application launch.

5.1.2 Throughput & Capacity

- **NFR-5:** The **Backend System** *shall* handle high-concurrency spikes of at least 5,000 simultaneous transactions during *flash reservations* without degrading API latency below 1000 milliseconds.
- **NFR-6:** Real-time push notifications *shall* be dispatched from the **Backend** within 2 seconds of a **Partner Business** publishing a new package.
- **NFR-7:** The **Database** *shall* support a connection pool capable of sustaining 10,000 concurrent read operations during peak business hours.

5.1.3 Resource Utilization

- **NFR-8:** The **Consumer Application** *should* consume no more than 100 MB of active RAM to ensure smooth operation on lower-tier mobile devices.
- **NFR-9:** The **Mobile Application** *shall* utilize OS-native *geofencing* APIs rather than continuous GPS polling to minimize device battery consumption.

Rationale:

- Strict response times prevent users from missing limited-time offers. High throughput guarantees platform stability. Optimized resource utilization ensures the app remains lightweight and user-friendly, preventing uninstalls due to battery drain.

5.2 Safety Requirements

The safety requirements for Süprüz focus on preventing catastrophic data loss, ensuring the physical well-being of users during the pickup process, and maintaining system integrity during unexpected failures.

5.2.1 Data Resilience & Fault Tolerance

- **NFR-10:** The **System** *shall* perform automated, encrypted full database backups every 12 hours to prevent critical data loss.
- **NFR-11:** The **Database** *shall* maintain write-ahead logs (*WAL*) to guarantee point-in-time recovery capabilities up to the last 5 minutes.
- **NFR-12:** The **System** *shall* automatically failover to a secondary instance with a Maximum Tolerable Downtime (*MTD*) of 30 minutes in the event of a critical hardware failure.

5.2.2 Graceful Degradation & Physical Safety

- **NFR-13:** In the event of a mapping service API outage, the **Mobile Application** *shall* gracefully degrade to a list-only view to maintain core functionality.
- **NFR-14:** The **Consumer Application** *should* display a standard safety disclaimer advising users to be aware of their surroundings when utilizing third-party map integrations for navigation.

Rationale:

- Data resilience ensures operations are not permanently disrupted by hardware faults. Graceful degradation keeps the business running during third-party outages. Physical safety requirements protect users and reduce legal liabilities.

5.3 Security Requirements

Security requirements define how the Süprüz platform protects sensitive user data, prevents fraudulent reservations, and ensures secure communication across all system components.

5.3.1 Authentication & Session Management

- **NFR-15:** The **System** *shall* utilize secure JSON Web Tokens (*JWT*) with a maximum expiration time of 24 hours for active mobile sessions.
- **NFR-16:** The **Backend System** *shall* immediately invalidate all active sessions upon a user-initiated password reset.

5.3.2 Threat Mitigation & Data Protection

- **NFR-18:** The **API Gateway** *shall* implement strict *Rate Limiting* (maximum 60 requests per minute per IP) to mitigate brute-force and DDoS attacks.
- **NFR-19:** The **Backend** *shall* sanitize all incoming user data to prevent *SQL Injection (SQLi)* and *Cross-Site Scripting (XSS)* vulnerabilities.
- **NFR-20:** All data exchanged across the network between the **Mobile Clients**, **Business Portal**, and **Backend** *shall* be encrypted in transit using TLS 1.2 or higher protocols.
- **NFR-21:** The generated **QR Code** *shall* be dynamically time-stamped and encrypted to prevent malicious actors from utilizing intercepted screenshots.

5.3.3 Transaction Integrity & Access Control

- **NFR-22:** The **Database** *shall* implement *pessimistic locking* mechanisms during the checkout process to prevent race conditions and double-bookings.
- **NFR-23:** The **Backend** *shall* enforce strict Role-Based Access Control (*RBAC*) to isolate permissions among **Consumers**, **Business Staff**, and **System Administrators**.

Rationale:

- Advanced threat mitigation protects the server architecture from malicious traffic. Strong cryptography and token management secure user identity. Transaction integrity ensures fairness and prevents fraudulent claims.

5.4 Software Quality Attributes

Software quality attributes define the architectural goals that make the Süprüz platform reliable, easy to use, and maintainable for the engineering team.

5.4.1 Usability & Portability

- **NFR-24:** The **Consumer Application** *shall* feature a frictionless UI, allowing a user to successfully reserve an available *Surprise Package* in no more than 3 screen taps from the main discovery map.
- **NFR-25:** The **Consumer Mobile Application** *shall* be fully functional and optimized for both iOS (version 15.0+) and Android (version 10.0+) operating systems.
- **NFR-26:** The **Business Portal** *shall* comply with standard web accessibility guidelines (*WCAG 2.1 Level AA*) to accommodate diverse users.

5.4.2 Maintainability, Testability & Scalability

- **NFR-27:** The **Backend Codebase** *shall* be packaged within **Docker** containers to ensure consistent deployment environments.

- **NFR-28:** The **System** *shall* maintain a minimum unit test coverage of 80% for all core domain logic.
- **NFR-29:** The system architecture *should* maintain strict domain boundaries to allow high-load modules to be seamlessly extracted into independent *microservices* in future scaling phases.

Rationale:

- High usability ensures high conversion rates. Emphasizing maintainability, testability, and containerization in the MVP phase reduces technical debt and allows rapid, bug-free iteration.

5.5 Business Rules

Business rules represent the core operational policies, restrictions, and conditions of the Süprüz platform. These rules dictate how the software must behave to satisfy the business model, maintain fairness, and protect both consumers and partner businesses.

5.5.1 Consumer Limits & Penalties

- **NFR-30:** A single **Consumer Account** *shall* be permitted to reserve a maximum of two (2) *Surprise Packages* per calendar day to prevent hoarding.
- **NFR-31:** The **System** *shall* automatically suspend reservation privileges for 7 days if a consumer fails to claim their package three (3) times within a 30-day period (defined as a *no-show*).
- **NFR-31b:** The system shall permanently suspend a consumer's reservation privileges upon receiving a second suspension within their account lifetime.
- **NFR-32: Consumers** *shall* be allowed to cancel a reservation without penalty only if the cancellation is submitted at least 60 minutes prior to the pickup window.
- **NFR-32b:** Refunds processed upon valid cancellation shall be reflected within 3 business days.

5.5.2 Merchant, Financial & Geolocation Policies

- **NFR-33: Partner Businesses** *shall* define a pickup time window of no less than 30 minutes when listing a *Surprise Package*.
- **NFR-34:** The listed price of any *Surprise Package shall* reflect a mandatory discount of at least 50% off the original retail value.
- **NFR-35:** The **System** *shall* restrict business hours input, preventing **Partner Businesses** from setting pickup windows that extend beyond their officially registered operating hours.
- **NFR-36:** The **System** *shall* validate that a user is physically located within a strict 5 KM radius of the business at the exact moment of reservation to enforce local commerce.

Rationale:

- Quotas and cancellation rules prevent platform abuse. Discount mandates maintain the platform's anti-waste identity. Geolocation enforcement protects the core business strategy of driving immediate local foot traffic.

6. Other Requirements

6.1 Localization and Internationalization
The Süprüz platform shall support Turkish (tr) and English (en). Users may select their preferred language from the profile settings screen. The application shall default to the device's system language upon first launch. Date, time, and currency formats shall adapt to the selected locale, with Turkish Lira (₺) as the default currency. Additional languages may be incorporated in future versions of the platform as needed.

6.2 Legal and Regulatory Requirements
The Süprüz platform shall comply with the Turkish Personal Data Protection Law (KVKK No. 6698) and the GDPR where applicable. Users must accept the Privacy Policy and Terms of Service during registration. The system shall collect only the data necessary for its core operations, and users retain the right to request permanent deletion of their personal data at any time (per FR-1.7).

Appendix A: Glossary

API (Application Programming Interface)
A set of rules and protocols that allows different software components to communicate with each other.

APNs (Apple Push Notification service)
A notification service provided by Apple for delivering push notifications to iOS devices.

Backend
The server-side component of the system responsible for data processing, business logic, and database interactions.

Business Partner / Merchant
A registered food business (e.g., restaurant, café, bakery, grocery store) that lists surplus food packages on Süprüz.

Client Application
The front-end mobile application used by consumers or business staff to interact with the Süprüz system.

Consumer
An end user who uses the Süprüz mobile application to discover, reserve, and pick up Surprise Packages.

CRUD (Create, Read, Update, Delete)
Basic operations used for managing data in a system.

Discovery Screen
The main interface in the consumer application where nearby businesses and available Surprise Packages are displayed via map or list view.

Docker
A containerization platform used to package applications and their dependencies for consistent deployment across environments.

FCM (Firebase Cloud Messaging)
A cross-platform messaging service used to send push notifications to Android devices.

Geofencing
A technology that uses GPS or location services to define a virtual geographic boundary, triggering actions when a user enters or exits that area.

GPS (Global Positioning System)
A satellite-based navigation system used to determine the real-time location of a device.

JWT (JSON Web Token)
A secure token format used for authentication and session management between client and server.

Map API (Mapping Application Programming Interface)
A third-party service (e.g., Google Maps, Mapbox) used to display maps, calculate distances, and provide geolocation features.

Modular Monolith
An architectural pattern where the system is built as a single deployable unit but internally divided into well-structured, independent modules.

No-Show
A situation where a consumer reserves a Surprise Package but fails to collect it within the designated pickup window.

OAuth 2.0
An authorization framework that allows users to log in using third-party services such as Google or Apple.

Package (Surprise Package)
A bundle of unsold food items offered by a business at a discounted price, with contents not fully specified in advance.

Pickup Window
The specific time interval during which a consumer must collect their reserved package from the business.

POS (Point of Sale)
The system used by businesses to process in-store payments.

QR Code (Quick Response Code)
A machine-readable code used in Süprüz to verify reservations during pickup.

RBAC (Role-Based Access Control)
A security model that restricts system access based on user roles (e.g., Consumer, Business Staff, Admin).

Reservation
The act of a consumer securing a Surprise Package through the Süprüz application.

REST (Representational State Transfer)
An architectural style for designing networked applications using standard HTTP methods.

SSO (Single Sign-On)
An authentication method allowing users to log in using a single set of credentials via third-party providers.

Surprise Package
A discounted food offering containing unsold items, where the exact contents may vary.

TLS (Transport Layer Security)
A cryptographic protocol used to secure data transmitted over the internet.

UI (User Interface)
The visual elements and layout through which users interact with the system.

UX (User Experience)
The overall experience and usability perceived by users when interacting with the application.

WAL (Write-Ahead Logging)
A database mechanism ensuring data integrity by recording changes before they are applied.

Appendix B: Analysis Models

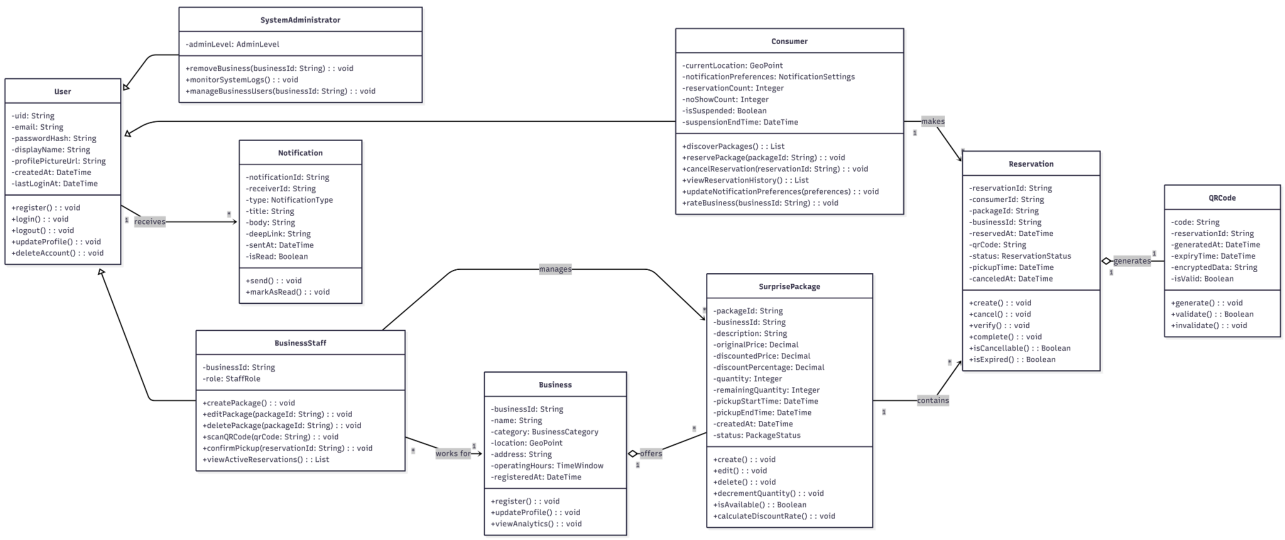

Figure 1: Hierarchical Class Diagram

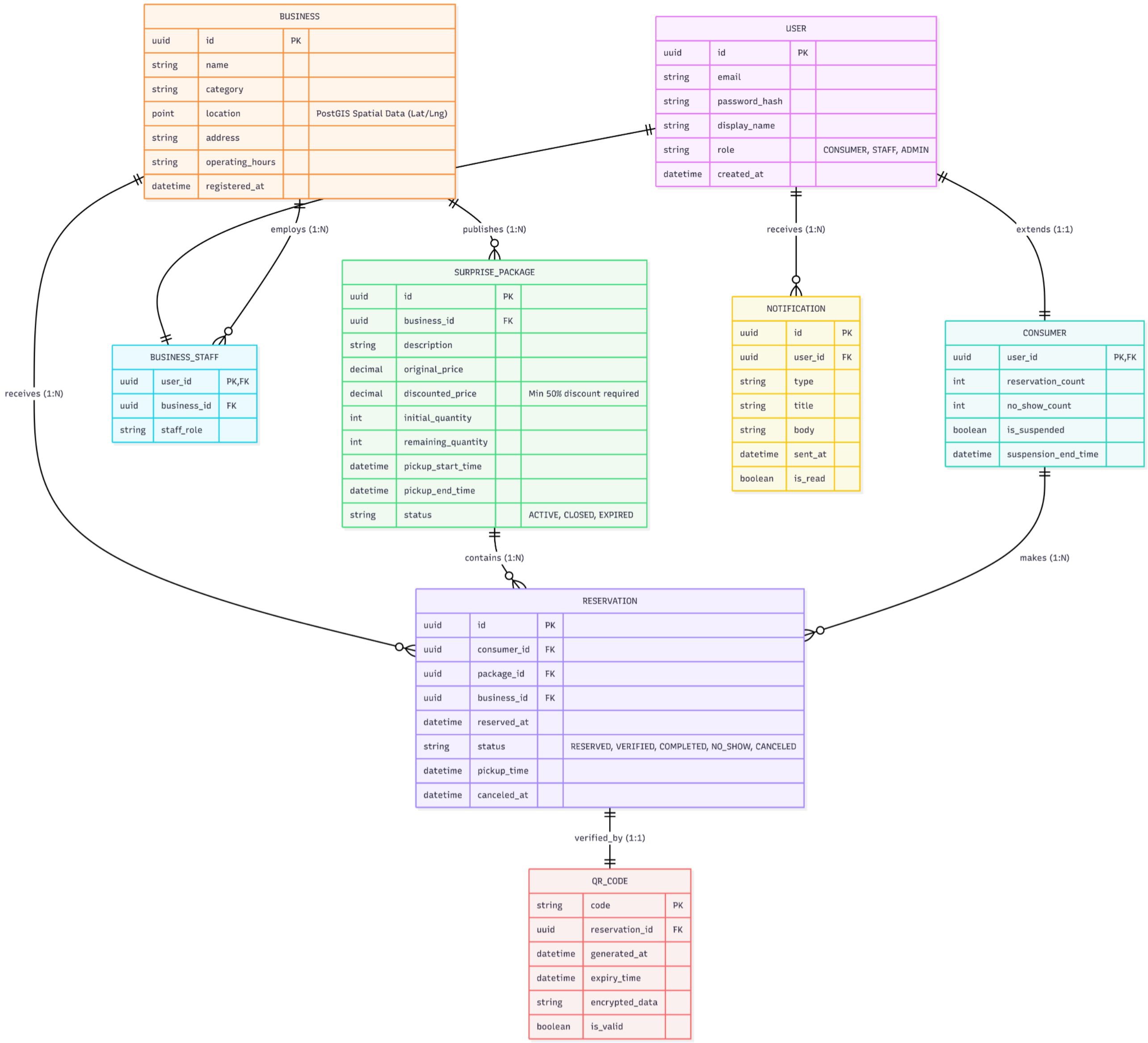

Figure 2: Crow Notation Entity Relationship Diagram

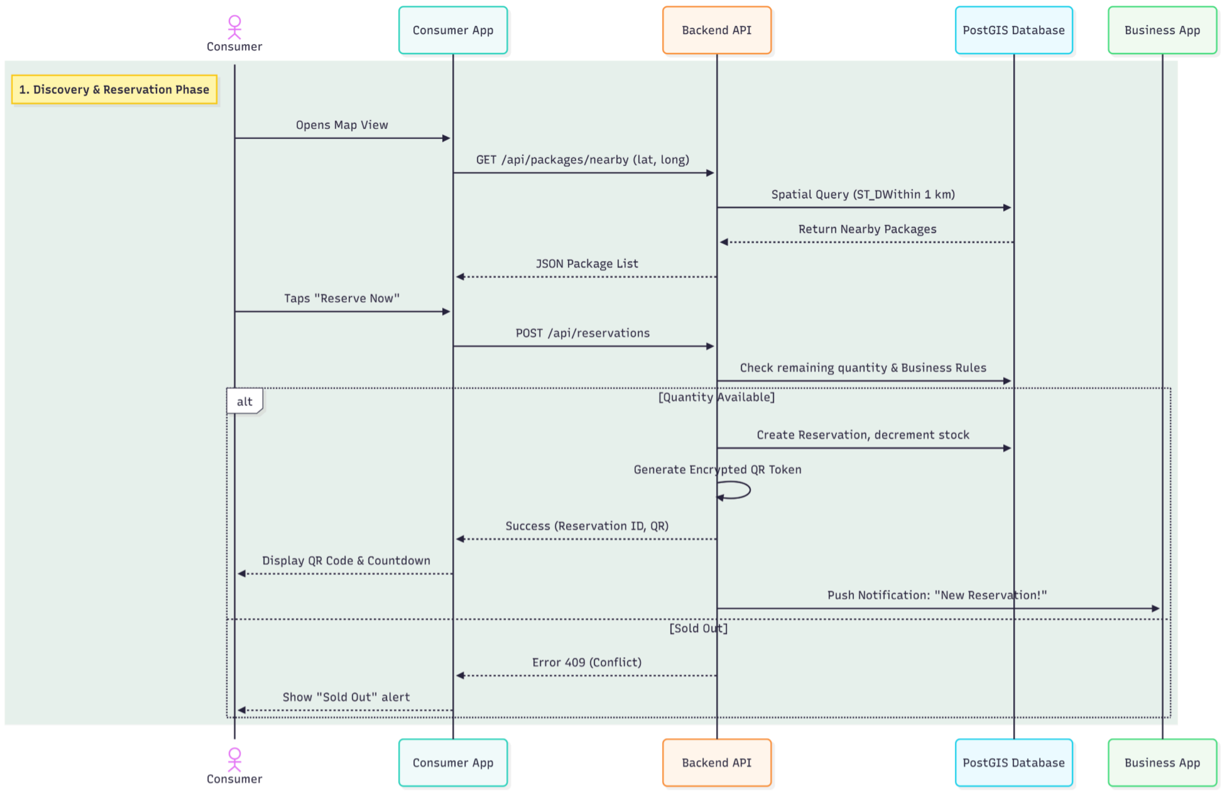

Figure 3: Sequence diagram illustrating the discovery of nearby packages and the reservation process.

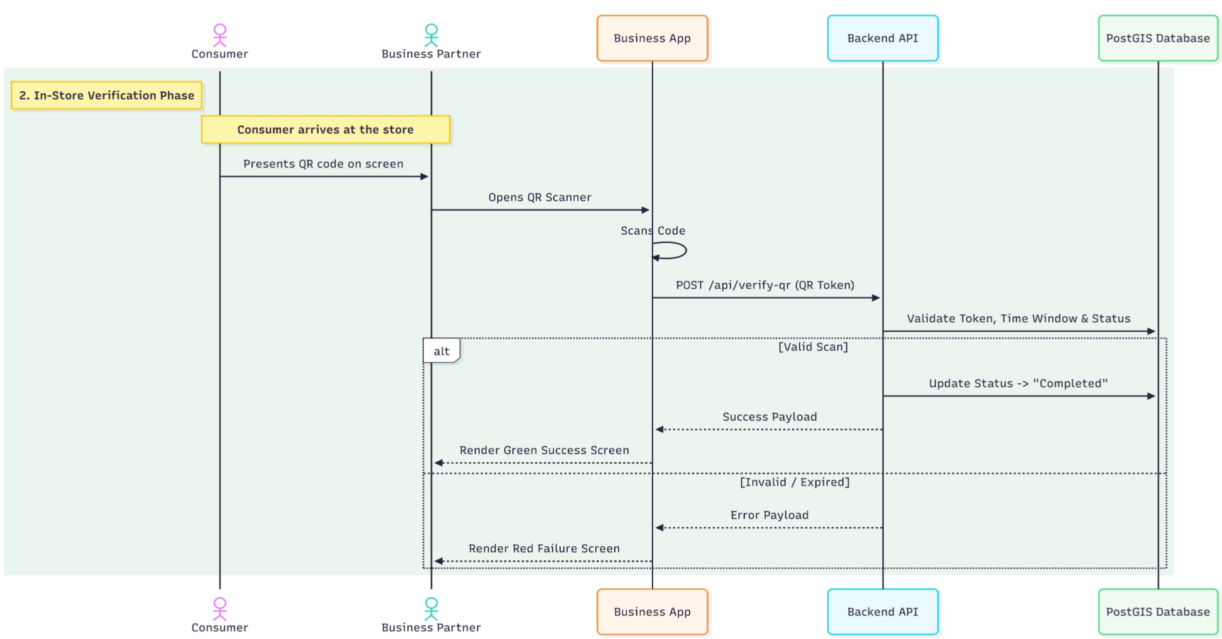

Figure 4: Sequence diagram detailing the in-store QR code verification and package pickup process.

Appendix C: Issues List

Note : Italics are solved issues.

Issue 1 *– No-Show Edge Cases Partially Resolved The no-show detection logic (FR-3.6) has been partially addressed. Business-side cancellation scenarios after consumer arrival are now covered under Section 2.6.5 (No-Show & Cancellation Policy Summary). The edge case of technical failures preventing QR scanning remains unresolved.*

Issue 2 – GPS Spoofing Mitigation TBD NFR-36 enforces the 5 KM proximity rule, however the specific server-side strategy for detecting and rejecting GPS spoofing attempts has not been formally defined.

Issue 3 – Notification Radius Not Specified *FR-6.2 references a configurable notification radius for new package alerts, but the default value and configuration mechanism have not yet been determined.*

Issue 4 – Offline Behavior Not Fully Specified *Section 4.2.3 mentions queuing non-critical actions upon reconnection, but the full scope of offline behavior across both the Consumer and Business applications remains unspecified*

Appendix D: Meetings List

Meeting 1 – March 1, 2026 – In-person

Initial team meeting conducted face-to-face to finalize the project idea, define the core concept of Süprüz, and align on the overall vision and objectives.

Meeting 2 – March 7, 2026 – Online

Follow-up session focused on task distribution and role assignments for the SRS document, determining which team member would be responsible for each section.

Meeting 3 – March 16, 2026 – Online

Technical coordination meeting in which the team adopted **Linear** as the project management and progress tracking system to monitor ongoing SRS development tasks.

Meeting 4 – March 19, 2026 – Online

Virtual session focused on design-related decisions, covering UI/UX expectations, visual direction, and interface considerations for the Süprüz platform.

Meeting 5 – March 24, 2026 – Online

Final review and refinement meeting addressing unresolved requirements, open issues from previous sessions, and overall finalization of the SRS document.